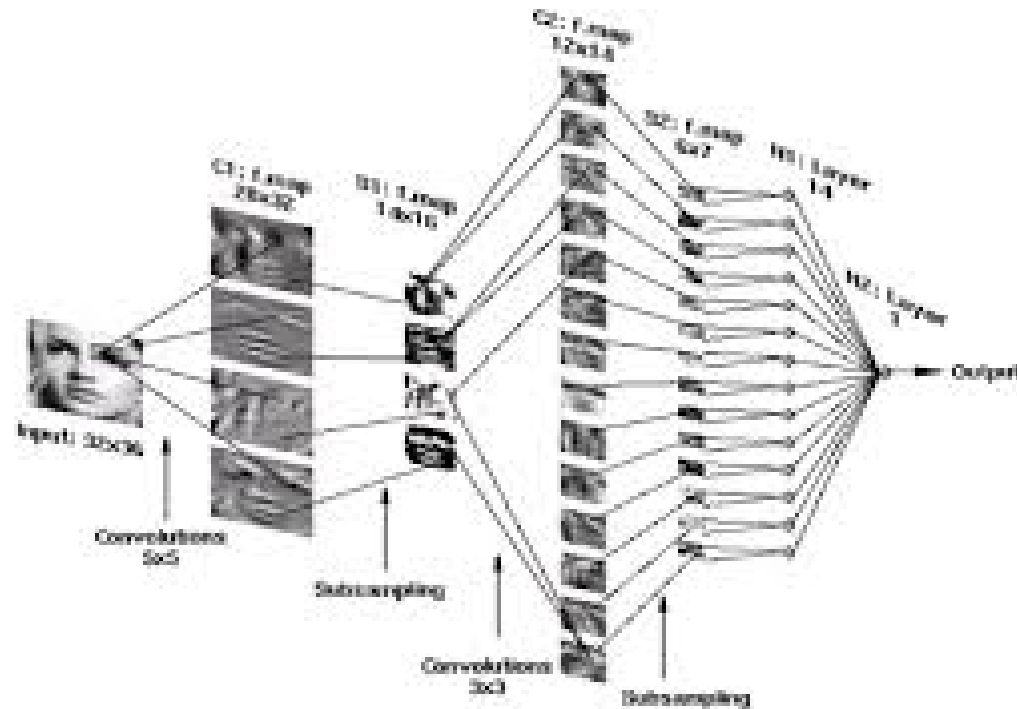


Mathematics of Machine Learning

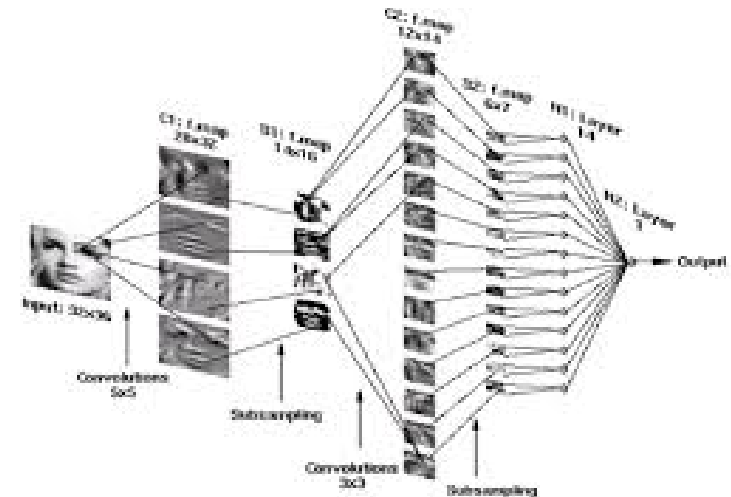
Introduction



Course Details and Topics

Outline

1. Why are you doing this course?
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers■
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation■
- Notes are on Moodle and
<https://ecs-vlc.github.io/aice3001/>■

Why are We Teaching You This?

- In a world where LLMs can do everything, why bother with learning mathematics?■
- You need to learn to trust your own judgement■
- LLMs can help us a lot, but if you don't trust yourself you are disempowered■
- This course is aimed at building up trust in your understanding of how machine learning works■

Why Mathematics?

- Learning machines are notoriously difficult to understand■
- Mathematics allows us to reason rigorously■
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition■—we can't cheat■that much■
- Mathematics also allow us to describe the world precisely■
- It also provides an arsenal of tools that turn out to be hugely useful■

Cracking the Code

- Mathematics is a language that takes time to master■
- For some it is easy, for others it will be a struggle■
- But given determination everyone can crack the code and understand what the mathematics is saying■
- This comes from engaging and working with maths■—this course has tutorials that should be considered compulsory■not because there are marks attached, but because without doing the work you aren't going to crack the code■

Starting from Different Places

- You all start from different places■
- Some of you may have a degree in mathematics■
- For others it may have been your least favourite subject■
- I invite strong mathematicians to become even stronger■—the best way to learn is through teaching others■
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others■

The Golden Rule of Mathematics

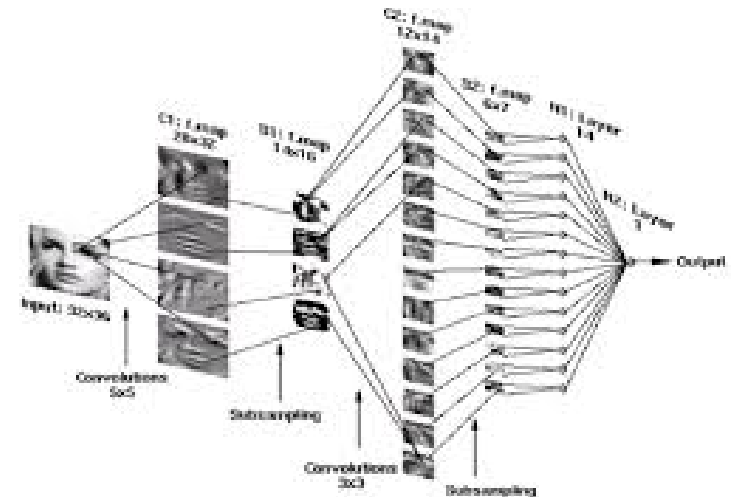
- Understand every step
- When you don't understand a step, stop, break the problem down, ask someone
- Mathematics uses a lot of short-cuts, which speeds-up calculations, but you need to be able to justify every step
- It takes bravery to admit you don't understand and insist that you will
- Asking “stupid” questions isn't stupid, it is how we all learn

The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning■
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas■
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides■
- The journey will pay you off■ these are the foundations of most of machine learning and will give you super-powers that are hard to acquire without mathematics■

Outline

1. Why are you doing this course?
2. **What we assume that you know**



What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni■
- You should know what a function is, e.g. $f(x) = x^2 + 3$ ■
- You should be able to do simple algebraic manipulations■
- And some elementary calculus

$$f'(x) = \frac{d(x^2 + 3)}{dx} = 2x$$

Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar■
- We will use bold symbols to represent **column vectors**, e.g.

$$\boldsymbol{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix} \blacksquare$$

- We denote **row vectors** as the **transpose** of column vectors
 $\boldsymbol{x}^T = (1.2, 2.3, 4.7)$ ■
- We care whether vectors are row or column vectors as it allows us to write equations involving vectors succinctly and unambiguously■

Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
- An $m \times n$ matrix acting on a vector in $\mathbb{R}^n$ transforms it to $\mathbb{R}^m$ vector, e.g.

$$\mathbf{M}\mathbf{x} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix} \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix} = \begin{pmatrix} 28.63 \\ -2.39 \end{pmatrix}$$

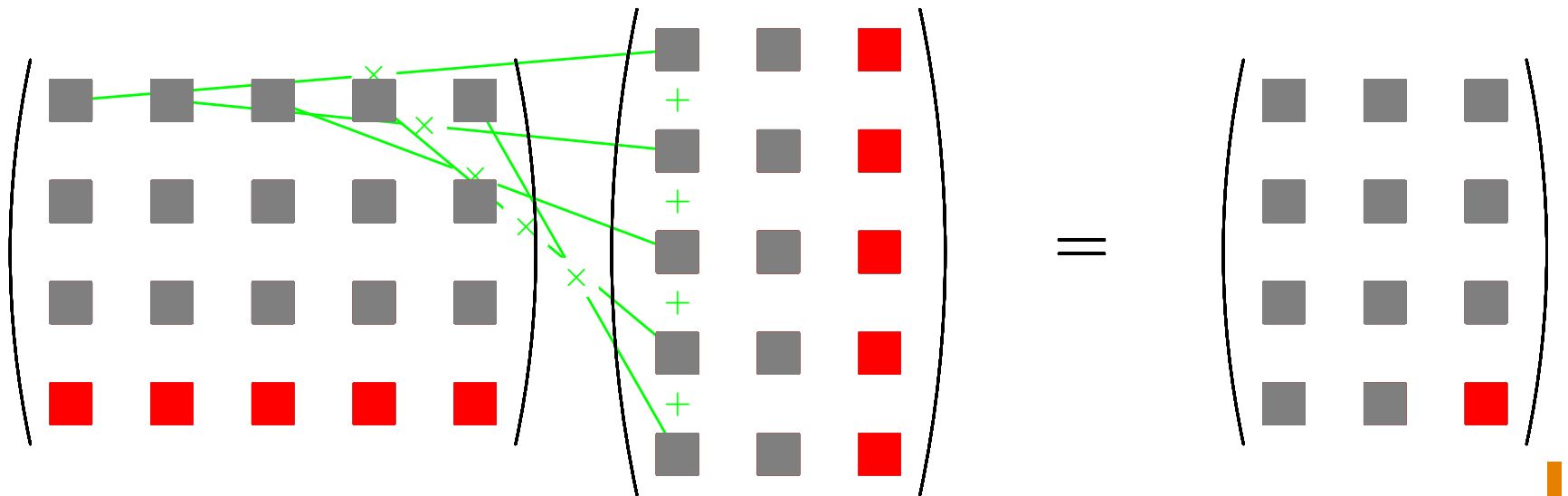
Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces■
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$ ■
 - ★ and then applying $\mathbf{A}$ to get vectors in $\mathbb{R}^\ell$

$$\mathbf{AB}x = \mathbf{A}(\mathbf{B}x) \blacksquare$$

Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)■



- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ ■
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$ ■

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\begin{aligned} \mathbf{a}^T \mathbf{b} &= (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a} \\ \mathbf{a}^T \mathbf{M} \mathbf{b} &= \mathbf{b}^T \mathbf{M}^T \mathbf{a} \end{aligned}$$

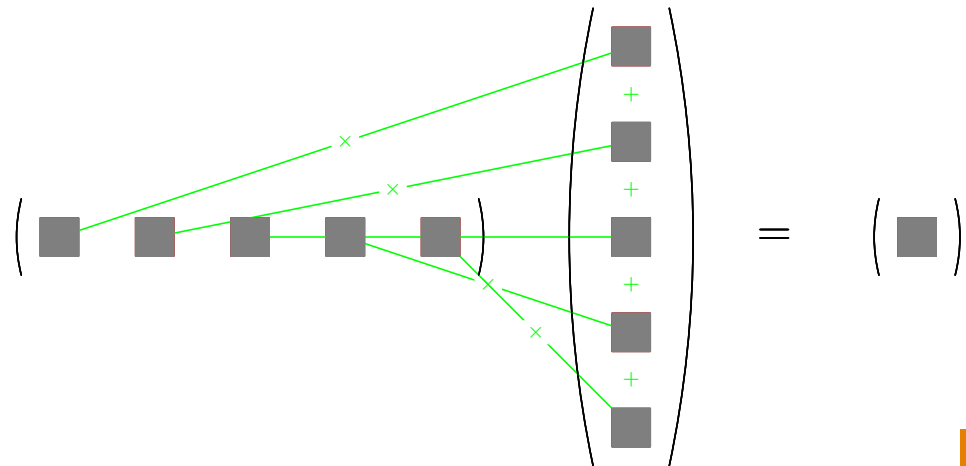
Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates the concept of treating a column vector as a matrix for multiplication. It shows a 4x5 matrix (represented by a grid of 20 squares, with the bottom row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). This is equated to a single 4x1 column vector (represented by a single column of 4 squares, with the bottom square highlighted in red). The entire equation is enclosed in large parentheses.

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $a^T b$

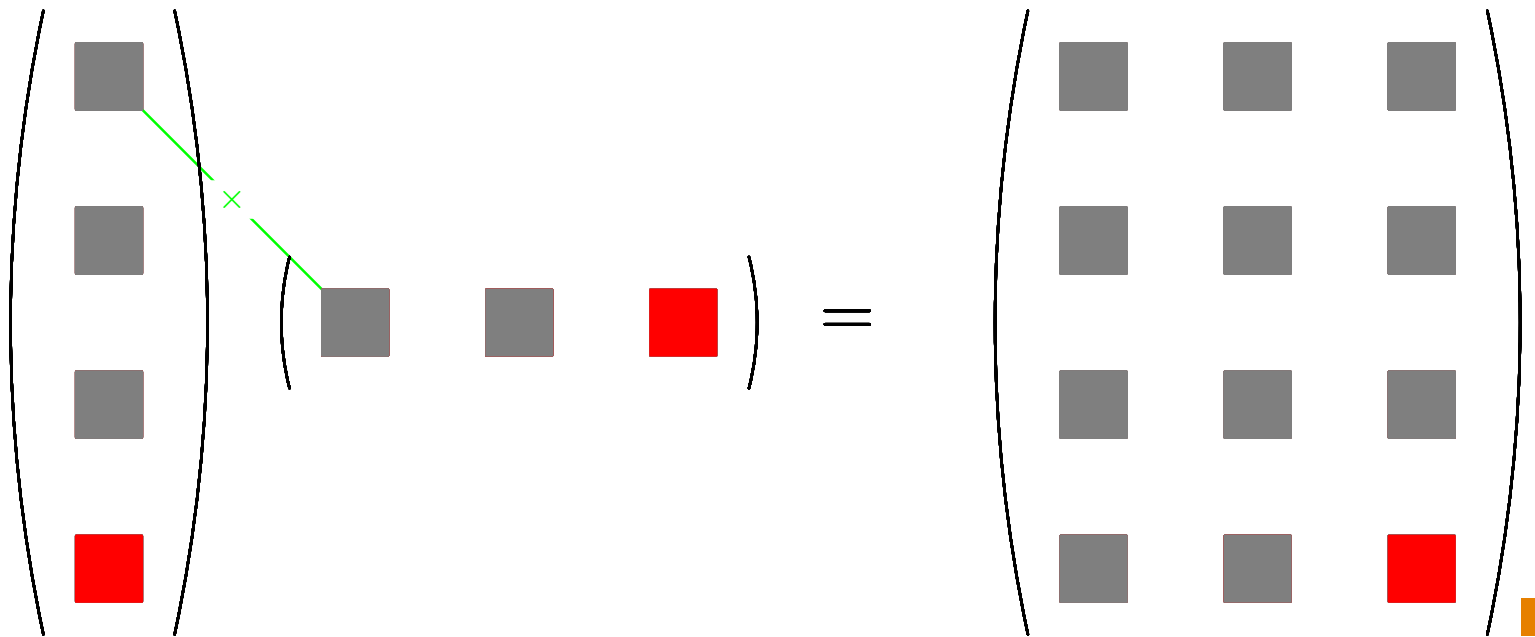


- Inner products encode angles between vectors

$$a^T b = \|a\| \|b\| \cos(\theta)$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T



- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij}a_j = M_{i1}a_1 + M_{i2}a_2 + \cdots + M_{in}a_n$$

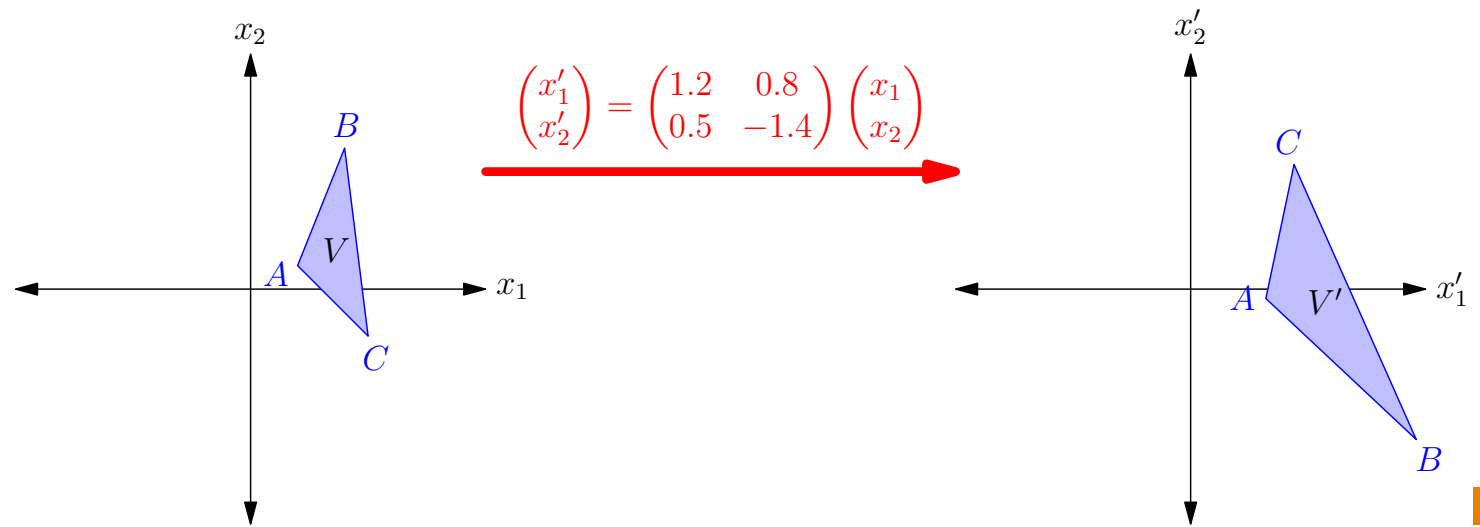
- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

Determinants

- A consequence of the fact that $\mathbf{AB}$ denotes the composition, apply mapping $\mathbf{B}$ then apply mapping $\mathbf{A}$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$$

- Determinants are only defined for square matrices, as non-square matrices project into different size spaces, where the volumes cannot be compared
- **Singular matrices** projects all points onto a lower dimensional space, so have determinant 0

Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$

- E.g. a learning machine for regression might be $\hat{f}(\mathbf{x}, \mathbf{w})$ or we might have a loss function $L(\mathbf{w})$

- We say a function f is linear if

$$f(c\mathbf{x}) = cf(\mathbf{x}) \qquad f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial f(\mathbf{x}^*)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}^*)}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x}^*)}{\partial x_n} \end{pmatrix}$$

- The gradient vector is orthogonal to the contour plane and points in the direction in which the function increases fastest

Probability

- We will have a lot to say about probability later in this course■
- To reason about events with uncertain outcomes we introduce **random variables**■
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event■
- E.g. we might assign 0 to the outcome of tossing a tail and 1 for a head■—this is an example of a discrete random variable■
- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park■

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x ■
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$ ■
- For continuous events the probability of X taking a particular value x is zero■
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$
■

Expectations

- One of the most useful quantities in dealing with random numbers is to look at the average value, aka as the **expectation**, for some function $g(X)$ of the random variable

$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) g(x) \\ \int f_X(x) g(x) dx \end{cases} \quad \blacksquare$$

- Expectations are linear operators

$$\mathbb{E}[cg(X)] = c\mathbb{E}[g(x)]$$

$$\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)] \quad \blacksquare$$

- The expectation of a constant is itself $\mathbb{E}[c] = c \quad \blacksquare$

Lesson

- I've covered material that we assume you know■
- If you don't, you are going to struggle following any machine learning course, so grab a maths book and revise this■
- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered■
- There are exercises we will go through in the tutorial session where you can test your proficiency■